

Flex Sensor Glove for Hand Motion Tracking in Physical Rehabilitation

Sofia G. Morais^{1,2}, Alexander Yevchenko¹, Riley E. Goodwin^{1,2}, Siera Gashi^{1,2}, Jim Mahaney¹
Dept. of Computer Science¹, Lampe Joint Dept. of Biomedical Engineering²
University of North Carolina at Chapel Hill



Background

Hand mobility is essential for daily function, yet it is often impaired by conditions such as Osteoarthritis, Carpal Tunnel Syndrome, tendon injuries, and post-fracture stiffness. Physical therapy is the primary method for restoring range of motion, however current practice relies on infrequent in-person evaluations, where therapists have limited visibility into whether patients are performing exercises correctly, progressing as expected, or even completing them at all. Standard measurement using manual goniometry has notable reliability issues, with errors of 21-31 degrees, while more accurate wearable systems are often prohibitively expensive.

This project aims to address these challenges by developing a low-cost wearable glove using flex sensors to continuously track key hand movements, improving accessibility and enabling better monitoring for both patients and therapists.

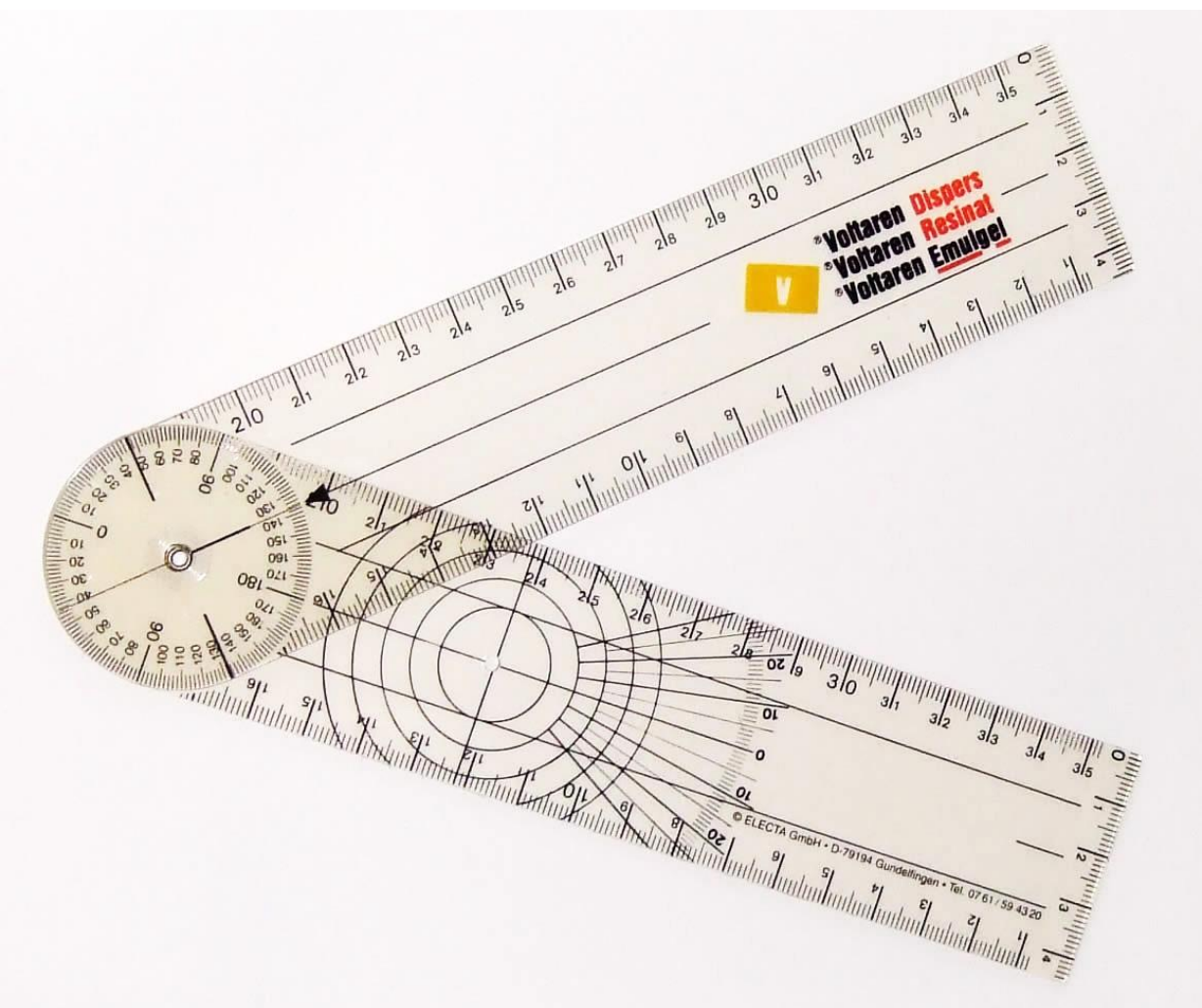


Figure 1. Goniometer

Methods

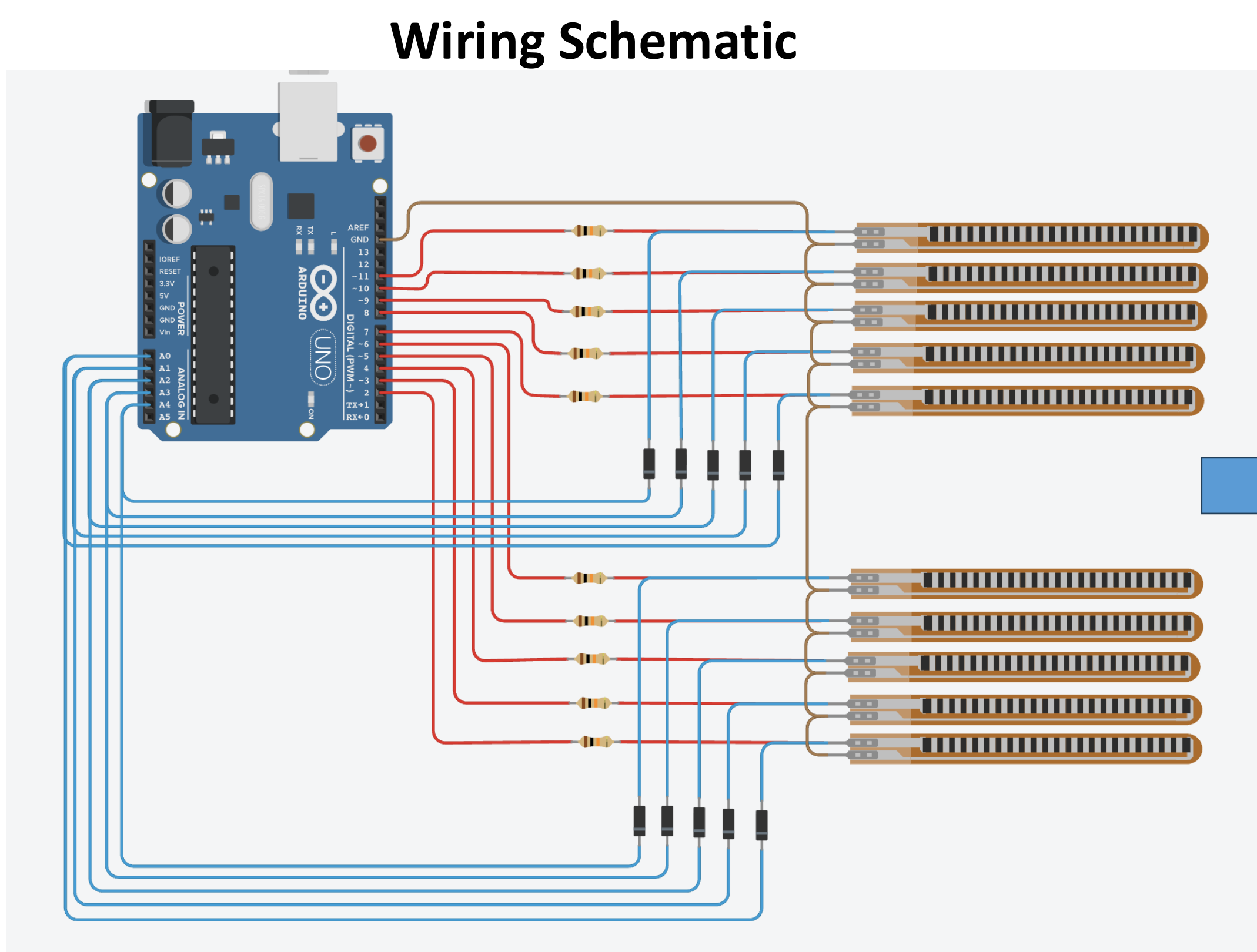


Figure 2. Wiring schematic on Tinkercad demonstrating how flex sensors are connected to digital vs. analog pins on Arduino board.

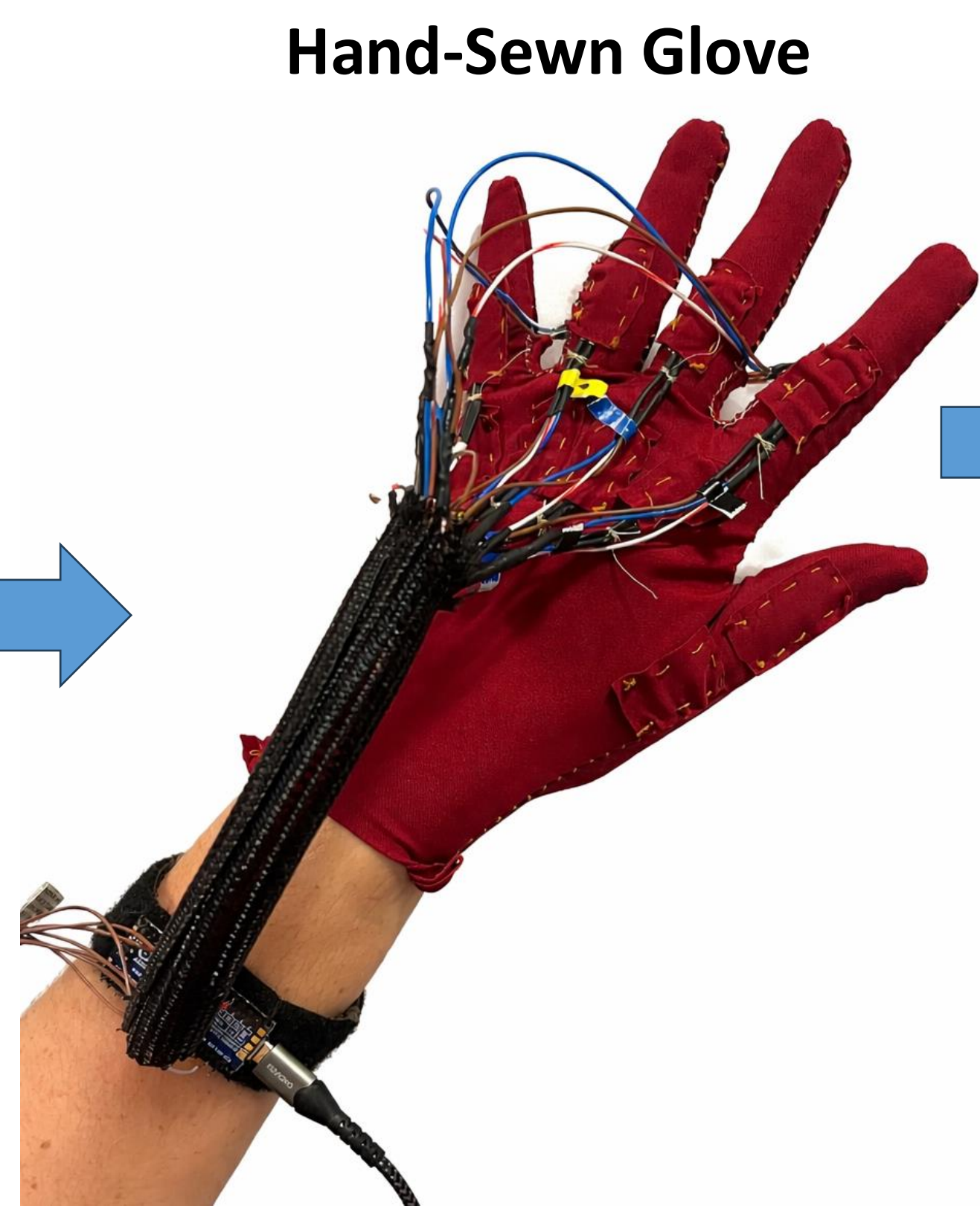


Figure 3. Hand-sewn glove with pockets for all bend and abduction sensors.

Arduino
99,761,344,324,211,228,155,206,1021,1021
99,749,343,324,211,227,152,206,1023,1021
100,736,343,324,210,228,149,205,1020,1021
99,724,343,325,210,229,151,206,1021,1018
100,712,343,322,211,229,152,206,1021,1020
99,703,343,322,208,227,151,205,1019,1020
99,695,342,323,210,226,151,207,1020,1020
100,685,343,323,211,228,153,207,1020,1021

Figure 4. On each row, an array of all sensor values prints into Arduino serial monitor.

MATLAB to analyze sensor-angle accuracy



Figure 5. Hand model in Unity provides live input of hand movement.

Flex sensors were integrated with an Arduino Nano33, and because the number of sensors exceeded the available analog inputs, a diode-based multiplexing approach was used to allow multiple sensors to share the same pins. The raw Arduino data was then sent to Unity, where both dynamic calibration and fixed min-max scaling were tested to accurately map sensor values to hand motion.

Results

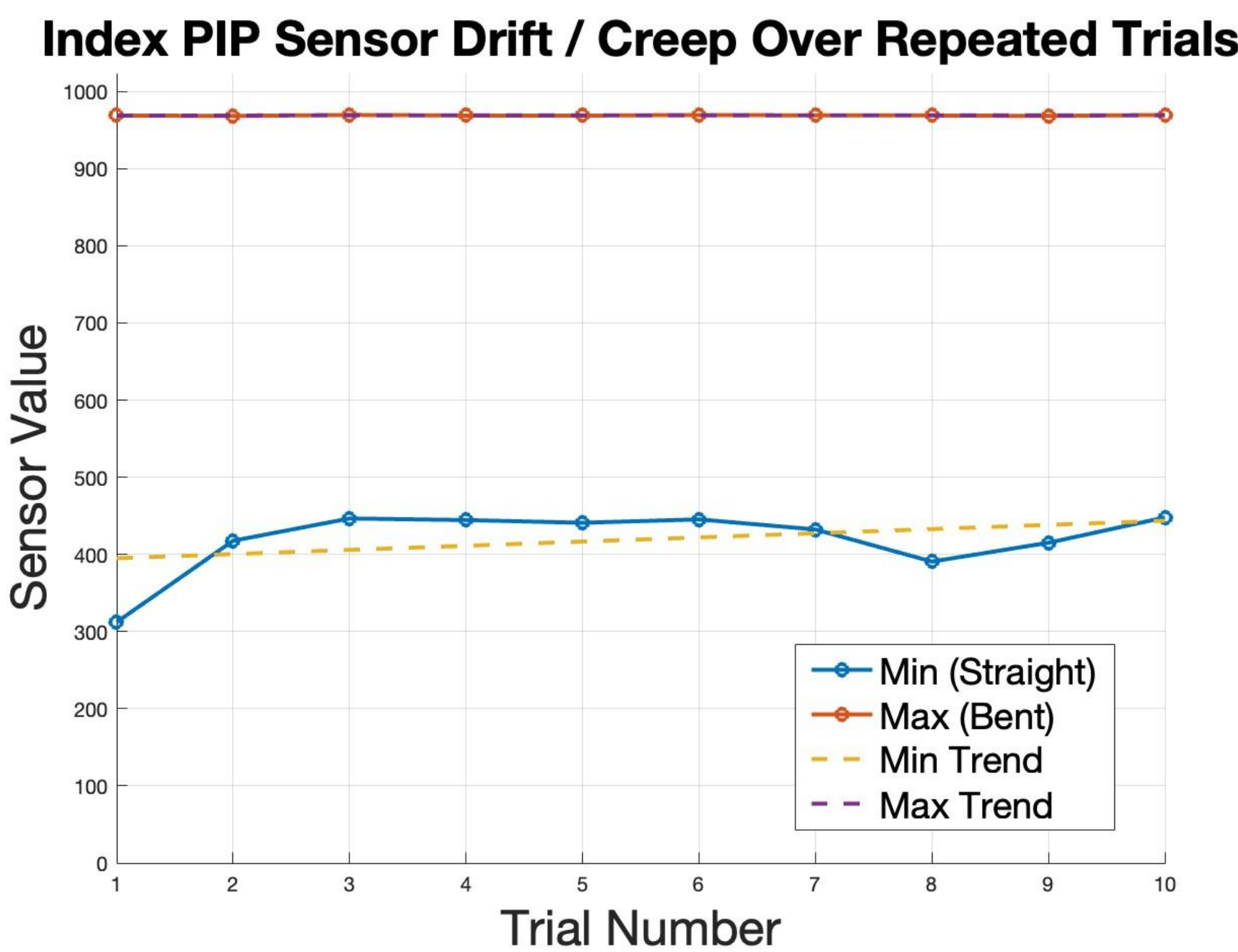


Figure 6. Measuring the maximum (bent) and minimum (straight) angles of the index proximal interphalangeal (PIP) joint across repeated trials.

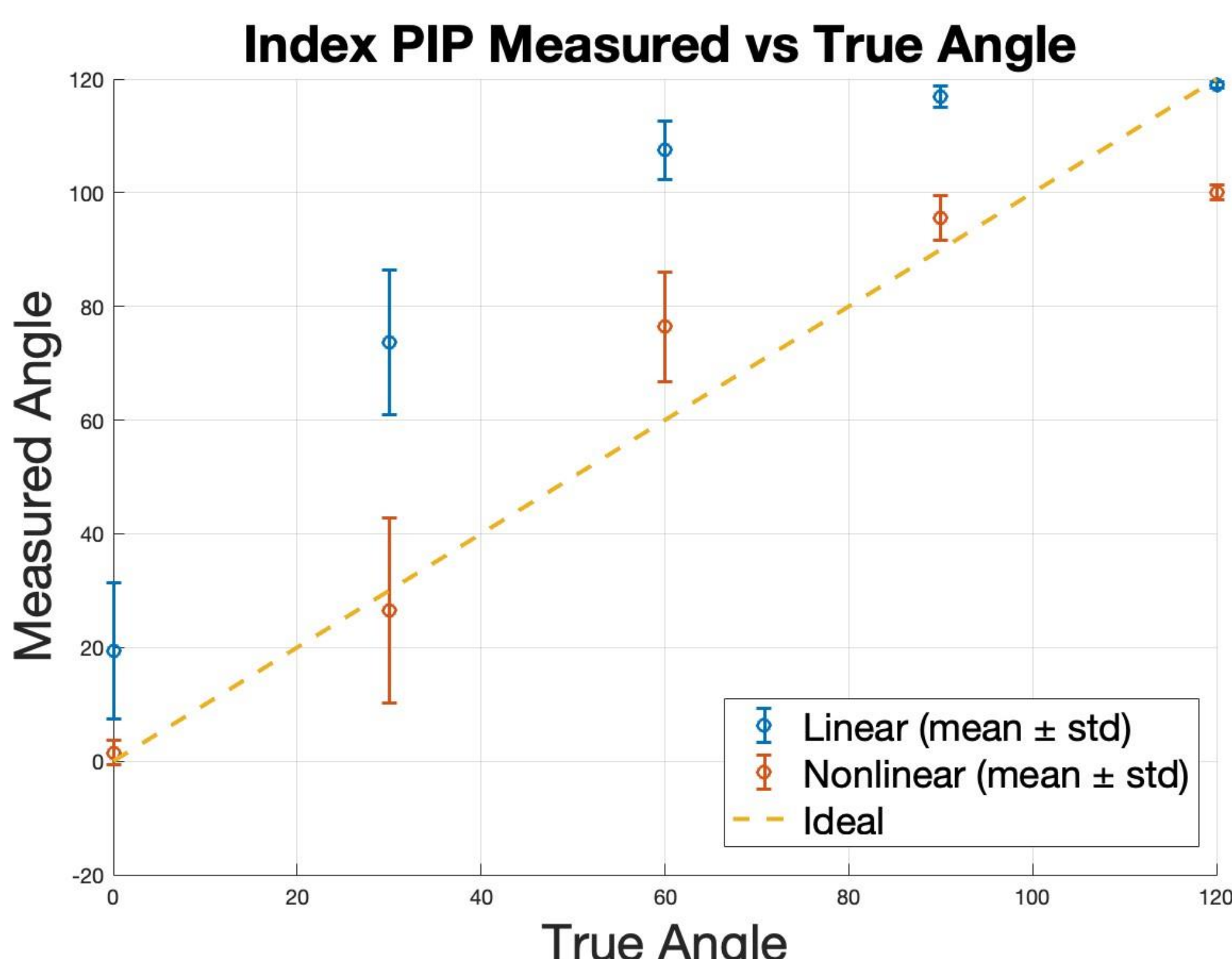


Figure 7. Comparing measured vs true angles at the index proximal interphalangeal (PIP) joint, with linear mapping and nonlinear calibration.

MATLAB plots highlight that while the maximum (bent) sensor values remained relatively stable across trials, the minimum (straight) values exhibited a slight upward trend, suggesting the presence of sensor drift / creep over repeated use. This indicates that baseline sensor values may shift over time and should be accounted for during calibration and angle mapping. Nonlinear calibration better captured the relationship between sensor output and joint angle, particularly in the mid-range where linear mapping overestimated the angles. However, both methods showed relatively large error bars, indicating variability in the measurements. Improved calibration methods and more consistent data collection would be necessary for reliable use in physical therapy applications.

Future Work

Design Improvements:

- Design a custom PCB to replace bulky wiring, making the glove more compact, durable, and easier to wear during repeated use.
- Add new types of sensors to capture thumb (carpometacarpal) abduction/adduction, a key movement often affected in hand osteoarthritis and not well measured by current low-cost flex sensors.^{1,2,5,6}

Physical Therapy Applications:

- Expand testing beyond a single healthy subject by including more participants and real exercise conditions and eventually evaluate the glove in patients with hand osteoarthritis to better understand its clinical accuracy.^{1,2}
- Use the glove in a structured home exercise program to track range of motion between clinic visits, giving therapists clear, session-by-session progress data that is not typically available at an affordable cost.^{3,4}

Acknowledgements / References

References

- [1] Kjekshus, et al. Limited effects of exercises in people with hand osteoarthritis: results from a randomized controlled trial. *Rheumatology*. 2018; <https://doi.org/10.1016/j.joca.2014.06.036>.
- [2] Jha B, et al. Measuring thumb range of motion in first carpometacarpal joint arthritis: The inter-rater reliability of the Kapandji Index versus goniometry. *J Hand Ther*. 2016; <https://doi.org/10.1177/1758998315616399>.
- [3] Shin HS, et al. Soft, skin-interfaced wireless electrogoniometry systems for continuous monitoring of finger and wrist joints. *Nat Commun*. 2025; <https://doi.org/10.1038/s41467-025-59619-z>.
- [4] Liu ZL, et al. The Effectiveness of Exercise-Based Rehabilitation in People With Hand Osteoarthritis: A Systematic Review With Meta-analysis. *J Orthop Sports Phys Ther*. 2024; <https://doi.org/10.2519/jospt.2024.12241>.
- [5] Caeiro-Rodríguez M, et al. A Systematic Review of Commercial Smart Gloves: Current Status and Applications/ Sensors. 2021; <https://doi.org/10.3390/s21082667>.
- [6] Indermun S, et al. Low-Cost Flexible Sensor Glove for Measuring Functional Finger Range of Motion in Daily Activities. In: *BIOSTEC*. Springer. 2024; https://doi.org/10.1007/978-3-031-96899-0_2.

Acknowledgements:

We are extremely grateful to Jim Mahaney and the Experimental Engineering Lab for the support, facilities, and guidance throughout the development of this project.